

PO0306

Ocular Brachytherapy with I-125 an Increasingly Important Option in Ocular Melanoma: A Case Series from a Low Middle Income Country, Latin American ExperienceJavier Rodriguez Corredor, Radiation Oncology Resident¹,David Antonio Martinez Perez, Radiation Oncology²¹Radiation Oncology, IMO Oncology Medical Institute, Neiva, Colombia; ²Radiation oncology, Las Américas AUNA Oncology Institute, Medellín, Colombia

Purpose: To evaluate the oncological results in local control and vision preservation at 5 years with ocular brachytherapy treatment with I-125 in the largest cohort of patients in Colombia and one of the largest in Latin America, given that the National Institute of Cancerology Bogotá-Colombia has an important recognition in being the only institution that performs this management in the country. In an emerging country with limited resources, where surgical management with exenteration for some neoplasms such as uveal melanomas is the treatment standard, ocular brachytherapy is becoming increasingly important.

Materials and Methods: This was a retrospective review of all patients with ocular melanomas treated with intraocular low-dose-rate (HDR) I-125 brachytherapy from July 2013 to October 2022 at the National Institutes of Medicine. cancerology, Bogotá-Colombia. 51 patients with ocular melanoma were identified. The median age at diagnosis was 53 years (range=21-62). The majority were choroidal melanomas (77%, n=39), ciliary body melanomas (19%, n=10), and two iris melanomas (4%, n=2). Average hospitalization during the procedure was 3 days. The most widely used apex radiation dose prescribed was 85Gy ($\pm$: 10.3 Gy) using the COMS plate. all patients had individualized planning based on CT, fused with fine-slice resonance, ultrasound.

Results: The mean follow-up for each patient varied from the first treated patients at 71.5 months ($\pm$: 51.8 months). 47 patients (92.1%) remained free of recurrence. 41 (80.3%) demonstrated complete tumor regression after a median period of 24 months (3-74 months). At the final examination, the visual acuity of 33 patients (64.7%) was 20/20 and 14 patients (27.4%) had a visual acuity of 20/40. Retinopathy was detected in 4 patients (7.8%). After the treatments, only two patients (3.9%) had active vascular changes at the last examination. Moderate optic neuropathy was observed in 1 patient (1.9%). Cataract development was diagnosed in 10 patients (19.6%) and 5 patients (9.8%) had bilateral cataract development.

Conclusions: Ocular brachytherapy with I-125 for ocular melanoma is an excellent treatment strategy for properly selected patients, offers optimal local control, preserves the visual, functional and aesthetic capacity of patients, our data on oncological results support this option every time more important in developing countries such as Colombia where access to this management modality is limited.

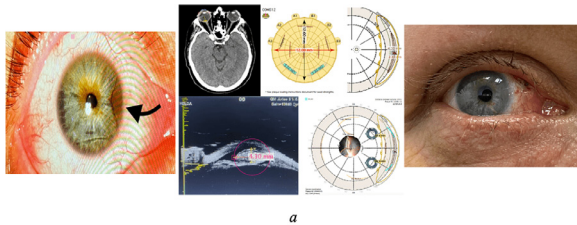


Figure 1. Patient with ocular iris melanoma, planning with computerized CT, ultrasound evaluating depth, planning and results with 85% regression of the lesion in 6 months. (a).

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Analysis of the Combined Treatment Efficacy of Radioactive Iodine-125 Seed Brachytherapy and Radiofrequency Ablation for Thoracic MalignanciesTao Zhu, MBBA¹, Zepaer Abudurehman, MBBA¹, Dengyao Liu, M.D., Ph.D.^{2,3}¹The Third Clinical Medical College of Xinjiang Medical University, Department of Interventional Radiology, Urumqi, China; ²Cancer Hospital Affiliated to Xinjiang Medical University, Xinjiang Key Laboratory of Translational Biomedical Engineering Research, Department of Interventional Radiology, Urumqi, Xinjiang Uygur Autonomous Region, China; ³Sun Yat-sen University Cancer Center, Department of Minimally Invasive Interventional Therapy, Guangzhou, China

Objective: To assess the combined treatment efficacy and safety of radioactive iodine-125 (¹²⁵I) seed brachytherapy and radiofrequency ablation (RFA) in patients with thoracic malignancies.

Materials and Methods: A retrospective analysis was conducted on 43 patients diagnosed with thoracic malignancies and treated at our facility. These patients were divided based on their treatment modalities into a control group of 20 patients, receiving only RFA, and a treatment group of 23 patients, receiving a combination of RFA and ¹²⁵I seed brachytherapy. We monitored tumor responses and complication rates at intervals of 1, 3, and 6 months post-procedure.

Results: Initial comparisons between the groups showed no significant differences in demographics such as age and sex, or baseline health metrics like preoperative hemoglobin levels ($P > 0.05$). Comparative analysis at 1, 3, and 6 months post-treatment also revealed no significant differences in overall response rates and local tumor control ($P > 0.05$ for all time points). However, the 6-month follow-up indicated a significant improvement in both the overall response rate and local control within the treatment group compared to the control group ($\chi^2=8.15$, $P < 0.05$). Importantly, the treatment was well-tolerated, with no reports of severe complications including internal bleeding, postoperative pain, or shock in either group. Minor complications, such as low-grade fever, were similarly infrequent and not significantly different between groups, occurring in 34.8% of the treatment group and 30% of the control group.

Conclusions: The integration of radioactive ¹²⁵I seed brachytherapy with RFA represents a minimally invasive, yet highly effective and safe therapeutic option for thoracic malignancies.

PO0308

Evaluating the Precision of 3D-Printing Template Technology in ¹²⁵I Seed Implantation for Lung Metastases: A Retrospective StudyDengyao Liu, M.D., Ph.D.^{1,2}, Tao Zhu, MBBA³, Zepaer Abudurehman, MBBA³¹Department of Interventional Radiology, Cancer Hospital Affiliated to Xinjiang Medical University, Xinjiang Key Laboratory of Translational Biomedical Engineering Research, Urumqi, Xinjiang Uygur Autonomous Region, China;²Department of Minimally Invasive Interventional Therapy, Sun Yat-sen University Cancer Center, Guangzhou, China;³Department of Interventional Radiology, The Third Clinical Medical College of Xinjiang Medical University, Urumqi, China

Objective: To evaluate the precision of three-dimensional printing template (3D-PT) technology for iodine-125 (¹²⁵I) seed implantation in treating lung metastases at the dosimetry level.

Methods: Between October 2022 and May 2023, 25 patients (17 males, 8 females; median age 61 years) with 28 lesions received ¹²⁵I seeds implantation assisted by 3D-PT in our hospital, as part of a retrospective